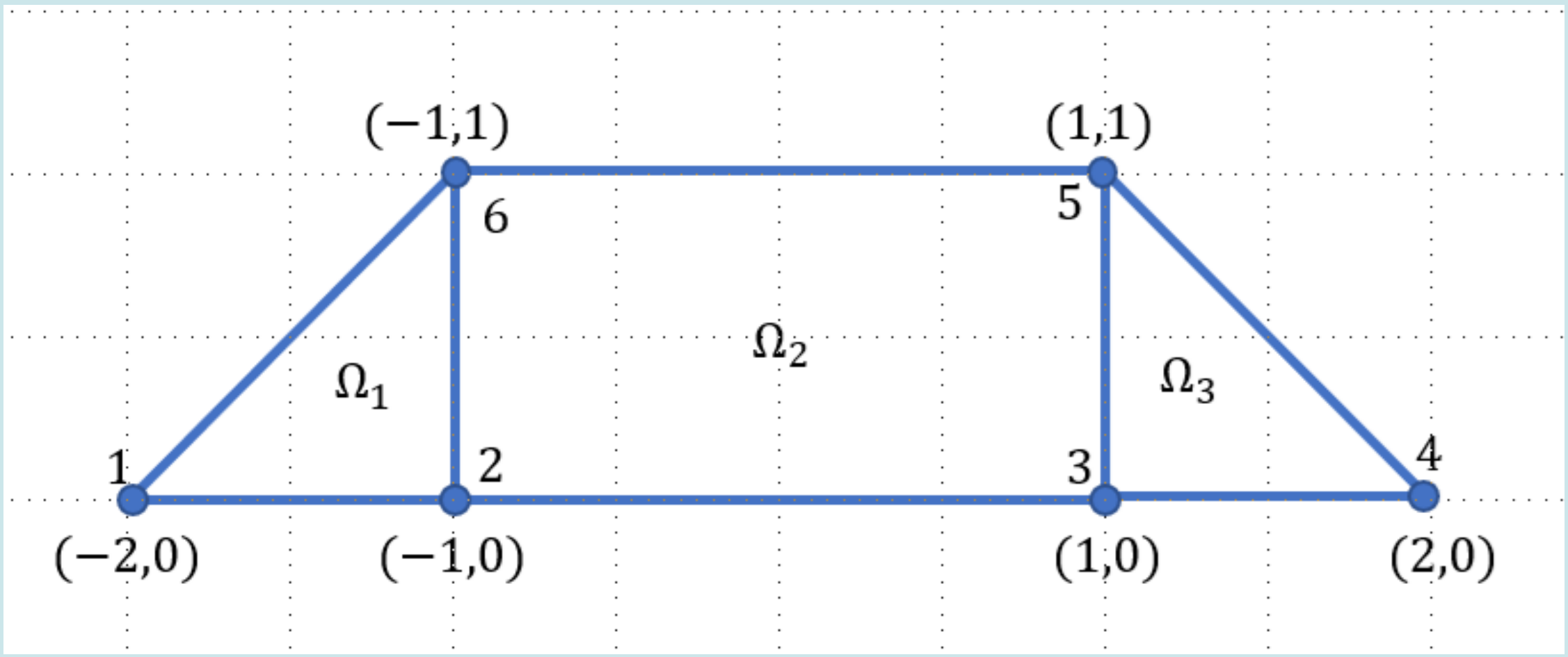
Fig-P2-Final-2Q-2024-25 Versió 3 (la més recent)

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0 Versió 1 (la més recent)

Pregunta 1

Resposta desada

Puntuat sobre 10,00

Consider the equation $-12\Delta u = 5$ on the domain Ω given in the figure. In all the different questions, the boundary condition on $y = 0$ is $u(x, 0) = 0$ and the two side edges given by 1-6 and 5-4 are insulated. Every question has different boundary conditions on $y = 1$. Using the mesh in three elements given by the connectivity matrix

$$C = \begin{pmatrix} 1 & 2 & 6 & * \\ 2 & 3 & 5 & 6 \\ 5 & 3 & 4 & * \end{pmatrix}$$

the system to solve becomes

$$\begin{bmatrix} 6 & -6 & 0 & 0 & 0 & 0 \\ -6 & 22 & 2 & 0 & -5 & -13 \\ 0 & 2 & 22 & -6 & -13 & -5 \\ 0 & 0 & -6 & 6 & 0 & 0 \\ 0 & -5 & -13 & 0 & 16 & 2 \\ 0 & -13 & -5 & 0 & 2 & 16 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \\ U_5 \\ U_6 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{bmatrix} + \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \end{bmatrix}$$

(a) (2 points) Which is the value of $F_1 \cdot F_6$?

- 2.88889e+00
- 3.22222e+00
- 2.55556e+00
- 2.66667e+00
- 3.11111e+00
- 3.00000e+00
- Empty answer (no penalty)
- 2.77778e+00

Hint: Recall that, for every element Ω^k, the vector F^k of a Poisson equation of the form $-c\Delta u = f_0$ has all the components equal to the constant f_0 times the area of the element divided by the number of nodes of Ω^k.

(b) (2 points) Assume $\frac{\partial u}{\partial y} = 0$ on $y = 1$. Which is the value of $u(1, 1)$?

- 3.51852e-01
- 2.59259e-01
- 9.42761e-02
- 1.26362e-01
- 2.73430e+00
- 1.48148e-01
- 2.88889e+00
- 2.51852e-01
- Empty answer (no penalty)
- 1.85185e-01
- 2.85470e+00

Hint: You can use that $U_5 = U_6$

(c) (3 points) Assume that $\frac{\partial u}{\partial y} = 1$ on $y = 1$. Which is the value of $u(1, 1)$?

- 1.22037e+00
- 6.45360e-01
- 7.56645e-01
- 1.18224e+00
- Empty answer (no penalty)
- 5.41342e-01
- 8.05484e-01
- 8.51852e-01

(d) (3 points) Assume that $\frac{\partial u}{\partial y} = \pi u$ on $y = 1$. Which is value of $u(1, 1)$?

- Empty answer (no penalty)
- -2.47767e-01
- -1.61633e-01
- -1.67963e-01
- -2.20143e-01
- -1.69212e-01
- -1.95841e-01
- -2.20155e-01

Hint: You can use that $U_5 = U_6$

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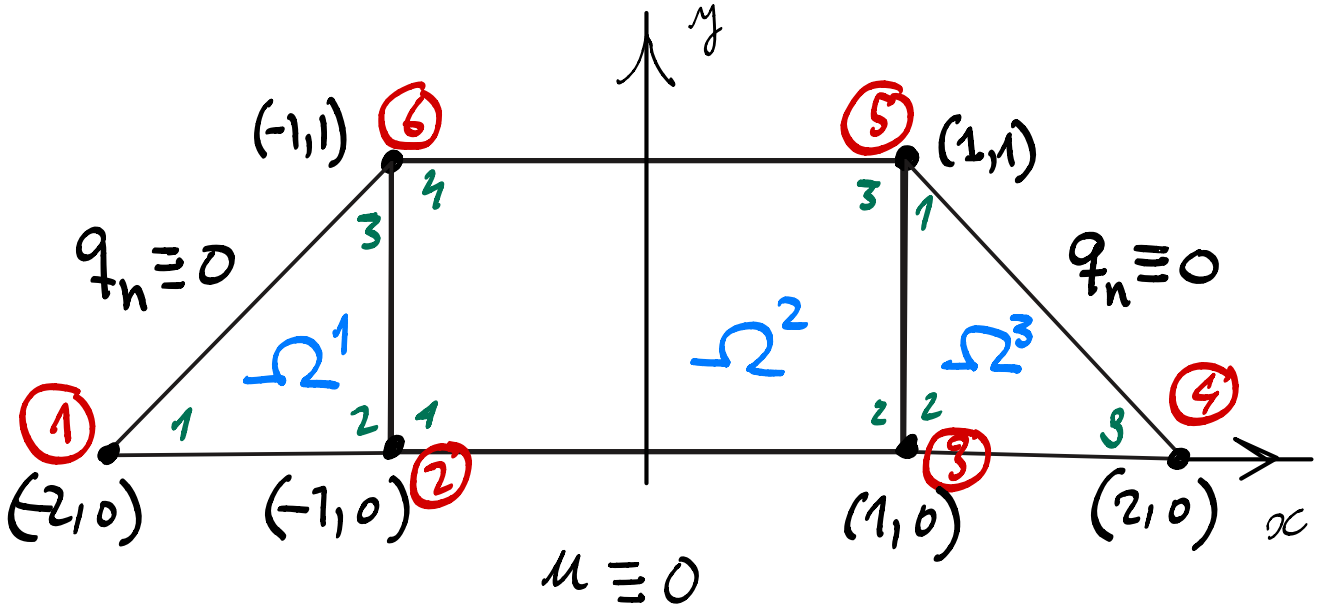
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$$-12\,\Delta u = 5$$

B.C. $y = 0,\ u(x,0) = 0,\ -2 \le x \le 2$



Connectivity matrix:

$$\mathcal{C} = \begin{pmatrix} 1 & 2 & 6 & * \\ 2 & 3 & 5 & 6 \\ 5 & 3 & 4 & * \end{pmatrix}$$

Coupled system

$$\begin{pmatrix} 6 & -6 & 0 & 0 & 0 & 0 \\ -6 & 22 & 2 & 0 & -5 & -13 \\ 0 & 2 & 22 & -6 & -13 & -5 \\ 0 & 0 & -6 & 6 & 0 & 0 \\ 0 & -5 & -13 & 0 & 16 & 2 \\ 0 & -13 & 5 & 0 & 2 & 16 \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \\ U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{pmatrix} + \begin{pmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \end{pmatrix}$$

(a) $F_1 \cdot F_6$? $F_i^e = \frac{f^e A^e}{n_e}$,

being n_e the number of nodes of element Ω^e

$$F_1 = F_1^1 = \frac{5 \cdot 1/2}{3} = \frac{5}{6}$$

$$F_6 = F_3^1 + F_4^2 = \frac{5 \cdot 1/2}{3} + \frac{5 \cdot 2}{4} = \frac{5}{6} + \frac{5}{2}$$

$$= \frac{5+15}{6} = \frac{10}{3}$$

So,

$$F_1 \cdot F_6 = \frac{5}{6} \cdot \frac{10}{3} = \boxed{\frac{25}{9} = 2.7}$$

(b) $\frac{\partial u}{\partial y} \equiv 0$, on $y = 1$, so $q^2_{n,3} \equiv 0$

fixed nodes: 1, 2, 3, 4

free nodes, 5, 6

$$Q_5 = Q^3_{1,3} + Q^2_{3,3} = 0,$$

and the reduced system to solve is,

$$\begin{pmatrix} 16 & 2 \\ 2 & 16 \end{pmatrix}\begin{pmatrix} U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} F_5 = 10/3 \\ F_6 = 10/3 \end{pmatrix}$$

$$- \begin{pmatrix} 0 & -5 & -13 & 0 \\ 0 & -13 & 5 & 0 \end{pmatrix}\begin{pmatrix} U_1 = 0 \\ U_2 = 0 \\ U_3 = 0 \\ U_4 = 0 \end{pmatrix}$$

$$+ \begin{pmatrix} Q_5 = 0 \\ Q_6 = 0 \end{pmatrix}$$

$$\Longleftrightarrow \begin{pmatrix} 16 & 2 \\ 2 & 16 \end{pmatrix}\begin{pmatrix} U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} 10/3 \\ 10/3 \end{pmatrix}$$

We can use Cramer's Rule (*)

$$\Delta = \begin{vmatrix} 16 & 2 \\ 2 & 16 \end{vmatrix} = 256 - 4 = 252$$

$$\Delta_5 = \begin{vmatrix} 10/3 & 2 \\ 10/3 & 16 \end{vmatrix} = \frac{10}{3}(16 - 2)$$

(*) Or, we can use that $U_6 = U_5$ (see Hint) to find U_5:

$$16U_5 + 2U_6 = 18U_5 = \frac{10}{3} \Longrightarrow U_5 = \frac{5}{27}$$

So: $U_5 = \dfrac{\Delta_5}{\Delta} = \dfrac{140/3}{252} = \dfrac{\cancel{2^2}\cdot 7\cdot 5}{\cancel{2^2}\cdot 3^3\cdot\cancel{7}} = \boxed{\dfrac{5}{27}}$

(c) $\dfrac{\partial u}{\partial y} \equiv 1$, on $y=1$

$$q^2_{n,3}(s) = \left\langle \begin{pmatrix} 12 & \\ & 12 \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial x}(1-s,1) \\ \frac{\partial u}{\partial y}(1-s,1) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$$

$$= \left(12\,\frac{\partial u}{\partial x}(1-s,1), 12\,\frac{\partial u}{\partial y}(1-s,1)\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= 12\,\frac{\partial u}{\partial y}(1-s,1) = 12, \quad 0 \le s \le 2$$

$$Q_5 = \overbrace{Q^3_{1,3}}^{=0} + Q^2_{3,3} = 0 + \frac{q^2_{n,3}\cdot h^2_3}{2}$$

$$= \frac{12\cdot\cancel{2}}{\cancel{2}} = 12.$$

And the corresponding reduced system is,

$$\begin{pmatrix} 16 & 2 \\ 2 & 16 \end{pmatrix}\begin{pmatrix} U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} 10/3 \\ 10/3 \end{pmatrix} + \begin{pmatrix} 12 \\ 12 \end{pmatrix} = \begin{pmatrix} 46/3 \\ 46/3 \end{pmatrix}$$

Remark 1. It's easily seen that $Q_6 = 12$ as well.

$$\Delta_S = \begin{vmatrix} 46/3 & 2 \\ 46/3 & 16 \end{vmatrix} = \frac{46}{3}(16-2) = \frac{7 \cdot 2^2 \cdot 23}{3}$$

$$U_S = \frac{7 \cdot 2^2 \cdot 23/3}{2^2 \cdot 3^2 7} = \frac{\cancel{4 \cdot 2^2} \cdot 23}{\cancel{2^2 \cdot 3^3 \cdot 7}} = \boxed{\frac{23}{27} = 0.851}$$

Remark 2. Note that, again $U_6 = U_S$, so we can use this to strightly find U_S.

(d) If $\frac{\partial u}{\partial y} = \pi u$ for $y = 1$, then

$$q_n(s) = 12 \frac{\partial u}{\partial y}(1-s, 1) = 12 \pi u(1-s, 1),$$

$0 \le s \le 2$, and this a convection B.C. with $\beta = -12\pi$, and $u_\infty = 0$. Therefore:

$$Q_S = \underbrace{Q^3_{1,3}}_{0} + Q^2_{3,3} = -\beta h_3 \left(\frac{U_S}{3} + \frac{U_6}{6} \right) + \frac{\beta h_3^2}{2} U_\infty$$

$$= \left\{ \begin{array}{l} \beta = -12\pi, \\ p_3^2 = 2, \\ U_\infty = 0. \end{array} \right\} = \frac{12\pi \times 2}{6} \left(2U_5 + U_6 \right)$$

$$= 4\pi \left(2U_5 + U_6 \right),$$

and

$$Q_6 = \underbrace{Q^1_{3,3}}_{0} + Q^2_{4,3} = -\beta p_3^2 \left(\frac{U_5}{6} + \frac{U_6}{3} \right) + \frac{\beta p_3^2}{2} U_\infty$$

$$= 4\pi \left(U_5 + 2U_6 \right);$$

so the reduced system now casts:

$$\begin{pmatrix} 16 & 2 \\ 2 & 16 \end{pmatrix} \begin{pmatrix} U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} 10/3 \\ 10/3 \end{pmatrix} + 4\pi \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} U_5 \\ U_6 \end{pmatrix}$$

$$\Longleftrightarrow \left[\begin{pmatrix} 16 & 2 \\ 2 & 16 \end{pmatrix} - 4\pi \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right] \begin{pmatrix} U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} 10/3 \\ 10/3 \end{pmatrix}$$

$$\Longleftrightarrow \begin{pmatrix} 8-4\pi & 1-2\pi \\ 1-2\pi & 8-4\pi \end{pmatrix} \begin{pmatrix} U_5 \\ U_6 \end{pmatrix} = \begin{pmatrix} 5/3 \\ 5/3 \end{pmatrix}. \quad (4)$$

Yet again (see Remark 2), it's easy to realize (see the Hint) that $U_5 = U_6$, and use it to solve the linear system (4).

This yields,

$$(8-4\pi)U_5 + (1-2\pi)U_6 \underset{U_5=U_6}{=} (8-4\pi+1-2\pi)U_5$$

$$= (9-6\pi)U_5 = 3(3-2\pi)U_5 = \frac{5}{3}$$

$$\Longrightarrow \boxed{U_5 = \frac{5}{9(3-2\pi)} = -1.69212 \times 10^{-1}}$$